SYPT 2027 · PROBLEM B8 "COLD DRINK" · THE STRONG CASE

Salt, water, ethanol and ice in an unstirred cooler: how cold does it get, and how fast does a part-filled can cool with no convection?

A closed-form model. You choose four masses: water, salt and ethanol at room temperature, and ice at 0 °C. Everything else is measured once or follows from an explicit formula.

§1 The system, the variables and the assumptions
§2 Part I · the bath in closed form: temperature, ice melted, composition, regimes, design rules
§3 Part II · the part-filled can, its air space, buoyancy, and where the ice and brine sit in the tank
§4 Part III · heat flow with no convection: the diluted brine, the slush, the cooling law
§5 The whole chain from your four masses to the drink temperature; design; what to measure
§6 Appendix: Laplace inversions, the classical Stefan limit, verification log, sources

The answer on one page

Headline numbers for your rig

Recipe (water / salt / ethanol / ice, kg)	T_b (°C)	t_5 (s): 25 → 5 °C			rate at 5 °C
		ice floats	lid board	ice at floor	(K/min, board)
B0 plain ice water (1.60 / 0 / 0 / 3.0)	0.0	643	640	640	0.22
B1 eutectic salt bath (1.60 / 1.00 / 0 / 4.0)	−21.2	223†	203	151	1.95
B2 salt + ethanol (1.00 / 1.00 / 0.30 / 4.0)	−24.4*	198	185	138	2.23
B3 ethanol only (1.00 / 0 / 0.80 / 4.0)	−13.2	332	324	222	1.06
B4 your original rig charge (2.60 / 0.95 / 0 / 3.0)	−14.7	477	447	287	0.81

Strict lower bound (no convection inside the can either), B1: $t_5 \approx 813$ s, rate at 5 °C ≈ 0.66 K/min.

Can 2 with 330 mL in your 14.2 × 29.0 × 17.5 cm tank, base 1 cm off the floor; drink well mixed; bath purely conductive. †at the brim (pack top 17.6 cm). *cotectic below −23.6 °C: extrapolated liquidus.

Six things the strong case says that a stirred model does not.

1. **Two closed forms carry everything.** The bath temperature is the positive root of a quadratic, Eq. (2.7). The drink temperature is $T_d(t) = T_b + \Theta_0\,[p_2\,\mathrm{erfcx}(p_2\sqrt{t}) - p_1\,\mathrm{erfcx}(p_1\sqrt{t})]/(p_2 - p_1)$, Eq. (4.10). It is not exponential (rate $\propto t^{-1/2}$ early, excess $\propto t^{-3/2}$ late).
2. **Your "coefficient lowered" appears by itself.** With nothing flowing, the meltwater from the ice next to the can cannot be carried away, so the brine there is diluted. The last ice beside the can melts at $T_L = -8.14$ °C, not at the bath's −21.2 °C. The no-convection coefficient is $\eta_c = \theta_L/\theta_b = 0.384$, Eq. (4.2); your mixing coefficient η_s (salt that never reached the ice) multiplies on top.
3. **The slush is still an excellent heat sink.** Its latent heat is spread through a partly melted ("mushy") zone, so for heat conduction it behaves like a material with effusivity $E \approx 2\,575$ W s$^{1/2}$ m^{-2} K^{-1}. That is 1.78 × the effusivity of clear brine (1 447).
4. **Where the ice sits matters more than ethanol.** Ice held down around the can: 151 s; floating ice leaves clear brine at the bottom: 223 s. Ethanol melts ice, so B3 is 1.59 × slower than B1.
5. **The air space matters less than you would think.** The drink's heat capacity and its wetted wall both scale with the fill, so almost doubling the drink (200→375 mL) moves t_5 only from 122 to 166 s. The air space matters for *buoyancy*: a 330 mL can floats even in fresh water and needs 1.11 N to hold it under brine.
6. **If even the drink cannot convect,** the wall jumps at once to the contact temperature $T_c = -7.38$ °C, below the drink's freezing point, and the drink takes about 813 s: the true minimum cooling rate,

4.0 × slower than well mixed.

How to read this

Every symbol is defined where it first appears. In the tables, C marks the four *control* variables, M a quantity you *measure* once (or take from a handbook), and D a *derived* quantity given by an explicit formula. The equation numbers in §5 chain them in order. Every number quoted is computed from those formulas. They were checked against the earlier numerical bath model over 116 recipes (worst error 0.73 K) and against a full numerical solution of the conduction equations (16 cases, rms error 4.4 %, worst 10.7 %).

1 The system, the variables and the assumptions

The experiment. An insulated tank (yours: glass, inner floor 14.2 × 29.0 cm, height 17.5 cm, in an EPS box). You tip in ice at 0 °C, then water, salt (NaCl) and ethanol at room temperature T_r, and wait for the bath to settle. Nothing stirs it. A can of drink at T_0 is lowered in and held upright with its base z_0 above the floor. Two questions:

(a) what temperature T_b does the bath settle at, and what is it made of (ice left, brine composition, solid salt left)?
(b) how does the drink temperature $T_d(t)$ fall when nothing moves except heat, by conduction?

The "strong case" combines everything at once: salt *and* water *and* ethanol with ice; no stirrer and no convection; the real can with its air space; and salt that is not perfectly mixed into the ice.

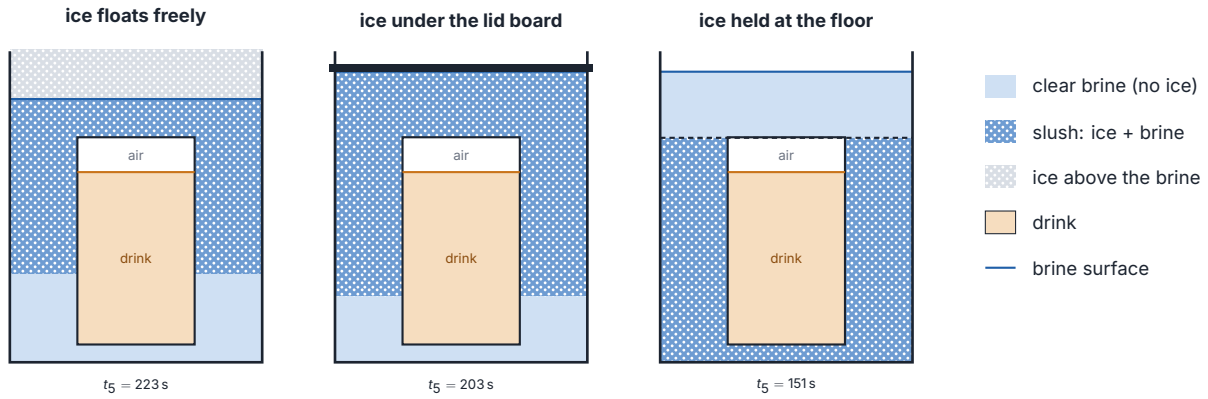


Figure 1. Your tank drawn to scale after recipe B1 has settled (4.0 kg ice, 1.60 kg water, 1.00 kg salt). Ice floats in brine, so where it ends up depends on how you hold it. Left: free-floating ice leaves 5.0 cm of clear (ice-free) brine at the bottom. Middle: ice pushed under a lid board. Right: ice held down on the floor by a mesh, with the clear brine on top. The can's drink (amber) fills 97.0 mm of its 116.6 mm height; the rest is air. Times are for 25 → 5 °C from §4.

1.1 The variables

Control variables (what you change between runs):

symbol	meaning	enters as
C m_w	mass of liquid water added, at T_r	sensible heat; solvent
C m_s	mass of NaCl added, at T_r	solute; dissolution heat
C m_e	mass of ethanol added, at T_r	solute; heat of mixing
C m_i	mass of ice added, at 0 °C	latent-heat store

Measured or tabulated once (M). Values are the ones used for every number in this document.

symbol	meaning	value	how to get it
thermodynamics			
c_w, c_i, c_s, c_e	specific heats: water, ice, NaCl (solid), pure ethanol (0–25 °C)	4 190, 2 060, 860, 2 345 J kg^{-1}K^{-1}	handbook
L_0	latent heat of fusion of ice at 0 °C	333.55 kJ/kg	handbook
$\Delta c = c_w - c_i$	makes the latent heat temperature dependent, $L(\theta) = L_0 - \Delta c\,\theta$	2 130 J kg^{-1}K^{-1}	derived
Δh_s	molar enthalpy of solution of NaCl (endothermic)	3.88 kJ/mol	handbook (placeholder)
$h_E(b_e)$	heat of mixing per mole of ethanol at 0 °C, $h_E = h_\infty + g\,b_e$	$h_\infty = -14.52$ kJ/mol, $g = 0.752$	fitted to H^E table
$c_{ea}(b_e)$	apparent specific heat of ethanol dissolved in water below 0 °C	$\min(6.31,\ 7.10 - 0.349\,b_e)$ kJ kg^{-1}K^{-1}	fitted to Melinder
M_s, M_e	molar masses of NaCl, ethanol	58.44, 46.07 g/mol	
liquidus (freezing-point) constants, fitted in §2.3			
K_s, ω_s	NaCl: depression constant and bound water per mole	3.173 K kg/mol, 0.0440 kg/mol	Bodnar (1993)
K_e, ω_e	ethanol: depression constant, bound water	2.116 K kg/mol, 0	Lange, Melinder
b_E, θ_E	eutectic brine molality and depression	5.163 mol/kg, 21.2 K	Bodnar (1993)
λ	drop in NaCl solubility per mol/kg of ethanol (along the cotectic)	0.0461 kg/mol	NIST ThermoML
mixing			
η_s, η_e	fraction of the salt / ethanol that actually reaches the liquid around the ice	1 (perfect)	§2.1: from T_b
slush, brine, drink, can, tank			
φ_p	volume fraction of ice in the packed slush	0.55	weigh ice in a known volume
ρ_i, k_i	ice density, conductivity	917 kg/m^3, 2.25 W m^{-1}K^{-1}	handbook
k_b, k_l	conductivity of cold brine; of the diluted melt next to the can	0.52, 0.52 W m^{-1}K^{-1}	handbook
ρ_0, β_s, β_e	brine density $\rho_b = \rho_0 + \beta_s w_s - \beta_e w_e$	1000, 765, 120 kg/m^3	weigh 100 mL of brine
ρ_d, c_d, k_d	drink density, specific heat, conductivity	1 040, 4 050, 0.56	10 wt% sugar solution
R, H, t_w, m_{Al}	can: outer radius, height, wall thickness, empty mass	33.00, 116.6, 0.10 mm; 13 g	callipers, balance
V_d	volume of drink in the can	330 mL	weigh full and empty
k_{Al}, c_{Al}	aluminium	205 W m^{-1}K^{-1}, 900 J kg^{-1}K^{-1}	handbook
A_t, H_t, z_0	tank floor area and height; height of the can base above the floor	412 cm^2, 17.5 cm, 1.0 cm	ruler
T_r, T_0, T_1	room temperature; drink start; target	25, 25, 5 °C	thermometer

Everything else is D *derived*. Each derived quantity is defined by one numbered equation, in the order in which you evaluate them (the full list is in §5).

1.2 Assumptions

A1 The bath reaches equilibrium before the can goes in (without a stirrer this takes a while; see A4). The tank exchanges no heat with the room during one run; a heat leak can be added as a known heat Q_x in Eq. (2.2).

A2 Heat capacities are constant within each step of the energy bookkeeping. The ethanol heat of mixing and apparent heat capacity depend on concentration and are updated once (a two-pass evaluation, §2.10).

A3 The ice liquidus of the mixed brine follows the Zdanovskii–Stokes–Robinson (ZSR) water-activity rule, with each binary liquidus written in a form fitted to data (§2.3).

A4 Imperfect mixing: only the fractions η_s and η_e of the salt and ethanol are dissolved in the liquid that touches the ice. The rest stays apart (salt crust on the floor, an ethanol layer on top).

A5 **No convection in the bath.** Heat moves only by conduction. Dissolved salt and ethanol do not move during a run: their diffusion is about 90 times slower than heat diffusion (§4.1).

A6 The drink is well mixed (one temperature T_d) in the main model. The limit in which the drink cannot

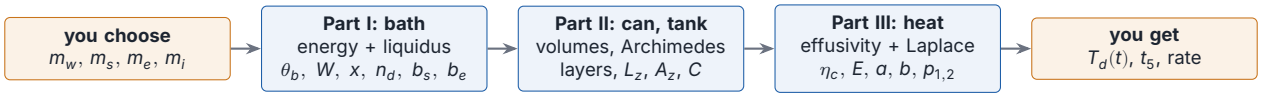
convect either is treated in §4.9.

A7 The can is a cylinder with a 0.1 mm aluminium wall, whose resistance ($t_w/k_{Al} \approx 5 \times 10^{-7}$ m^{2}K/W) is negligible. Its dome, neck and coatings are ignored. The air space conducts no heat.

A8 Ice packs with ice volume fraction φ_p, and brine fills the pores below the brine surface. The part of the can above the brine surface exchanges no heat (ice touching dry aluminium through air).

A9 On the time scale of one cooling the slush and brine around the can act as unbounded. The thermal disturbance reaches about 2 cm, less than the 3.8 cm clearance to the tank wall.

A10 The drink (about 10 wt% sugar) does not freeze in the main model. Its freezing point is −0.77 °C; see §4.9 for when the wall goes colder than that.



2 Part I · The bath in closed form

What decides the bath temperature

Two conditions must hold at the same time.

- **Energy.** The heat that the warm water, salt and ethanol bring in, plus the heat of solution and of mixing, has nowhere to go but into melting ice.
- **Phase equilibrium.** Ice can only sit in a liquid that is on its *liquidus*: the more dissolved solute per kilogram of liquid water, the colder that liquid must be.

Melting ice adds water and dilutes the brine, so the two conditions pull against each other. Solving them together gives one temperature.

Notation. Write the bath temperature as a depression below 0 °C, $\theta \equiv -T_b$ (in K, positive when colder than 0 °C). W is the mass of liquid water in the bath at the end, $x = W - m_w$ the mass of ice that melted, and n_d the moles of salt dissolved.

2.1 The mixing coefficient

With no stirrer, some of the salt you pour in never meets the ice: it sinks as a crust, or sits in a pocket of already saturated brine. Ethanol is lighter than brine and can float as a layer. Let the fractions that do end up in the liquid touching the ice be η_s and η_e (between 0 and 1). The *active* amounts are

$$n_s' = \eta_s \frac{m_s}{M_s}, \qquad n_e' = \eta_e \frac{m_e}{M_e}. \tag{2.1}$$

This is the "lowered coefficient" in its simplest form. In every formula below only n_s' and n_e' lower the freezing point and absorb or release heat of solution, so the effective depression constant per kilogram of salt you added drops from K_s/M_s to $\eta_s K_s/M_s$. The inactive salt still has to be cooled, so it stays in the heat capacities. How to measure η_s is in §2.9.

2.2 Energy balance

Enthalpy is a function of state, so any imaginary path from the start to the end gives the same answer. Take this one:

(i) cool the water, salt and ethanol from T_r to 0 °C, releasing $C_r\,\Delta T_r$;

(ii) at 0 °C, melt x of ice (absorbs xL_0), dissolve n_d moles of salt (absorbs $n_d \Delta h_s$), and mix n_e' moles of ethanol into the water (absorbs $n_e' h_E$; $h_E < 0$, so it actually releases heat);

(iii) cool everything from 0 °C to $-\theta$, releasing $\theta\,[\,(m_w + x)c_w + (m_i - x)c_i + m_s c_s + \eta_e m_e c_{ea} + (1-\eta_e)m_e c_e\,]$.

In an insulated tank, heat released equals heat absorbed. Collecting the terms in x:

$$\boxed{x\,(L_0 - \Delta c\,\theta) = C_r\,\Delta T_r + C_\theta\,\theta - n_d\,\Delta h_s - n'_e\,h_E + Q_x} \tag{2.2}$$

$$C_r = m_w c_w + m_s c_s + m_e c_e, \qquad C_\theta = m_w c_w + m_s c_s + \eta_e m_e c_{ea} + (1-\eta_e) m_e c_e + m_i c_i, \qquad \Delta T_r = T_r - 0\,^\circ\mathrm{C}. \tag{2.3}$$

Reading Eq. (2.2) term by term:

- **left side:** the heat needed to melt x of ice at the bath temperature. $L_0 - \Delta c\,\theta$ is the latent heat at $-\theta$ (Kirchhoff's law): 333.55 kJ/kg at 0 °C, about 13 % less at the eutectic.
- $C_r\Delta T_r$**:** the heat the warm inputs bring.
- $C_\theta\theta$**:** the heat released as everything cools below 0 °C. This includes the ice, which starts at 0 °C and must be cooled too.
- $n_d\Delta h_s$**:** the heat dissolution of salt takes.
- $-n'_e h_E$**:** the heat mixing ethanol gives.
- Q_x**:** any extra heat in (heat leak, or the can's heat; 0 for the bath alone).

2.3 The liquidus with bound water

Why not Blagden's law. $\theta = iK_f b$ with $i = 2$ assumes an ideal, dilute solution. Against Bodnar's (1993) measured NaCl liquidus it is wrong by up to 2.0 K: too cold below about 16 wt% and too warm above, because the ions are not ideal.

A form that fits and keeps the algebra closed. Suppose each mole of solute ties up ω kilograms of water in its hydration shell, so that only $W - \omega n$ of the liquid water acts as free solvent. A dilute-law depression on the free water then reads

$$\theta = \frac{K\,n}{W - \omega\,n} \quad\Longleftrightarrow\quad \theta = \frac{K\,b}{1 - \omega\,b}, \qquad b = \frac{n}{W}. \tag{2.4}$$

Fitted to Bodnar's liquidus and forced through the eutectic ($b_E = 5.163$ mol/kg at $\theta_E = 21.2$ K), NaCl gives $K_s = 3.173$ K kg/mol and $\omega_s = 0.0440$ kg/mol. That is 2.44 water molecules per NaCl, and the worst error from 0 to -21.2 °C is 0.05 K, against 2.0 K for Blagden. For ethanol the fitted ω_e comes out as zero (it is not an ion), leaving $\theta = K_e b_e$ with $K_e = 2.116$ K kg/mol (worst error 0.77 K over 0–28 K; Fig. 2).

Mixing the two (ZSR). The ZSR rule says a mixed brine freezes at the temperature where each solute's share of "water activity" adds up. If $b_{j,0}(\theta)$ is the molality of solute j alone that freezes at θ, then $\sum_j b_j / b_{j,0}(\theta) = 1$. Inverting Eq. (2.4) gives $b_{j,0} = \theta/(K_j + \omega_j\theta)$, so

$$\frac{b_s(K_s + \omega_s\theta) + b_e(K_e + \omega_e\theta)}{\theta} = 1 \quad\Longrightarrow\quad \theta = \frac{K_s b_s + K_e b_e}{1 - \omega_s b_s - \omega_e b_e}.$$

With $b = n/W$:

$$\boxed{\theta\,(W - W_b) = N, \qquad N = K_s\,n'_s + K_e\,n'_e, \qquad W_b = \omega_s\,n'_s + \omega_e\,n'_e} \tag{2.5}$$

N (in K kg) is the "depression strength" of what is dissolved. W_b is the bound water. Against the numerical ZSR model built on the full data tables, Eq. (2.5) is within 1.31 K everywhere (rms 0.36 K), and the worst case is at -30 °C.

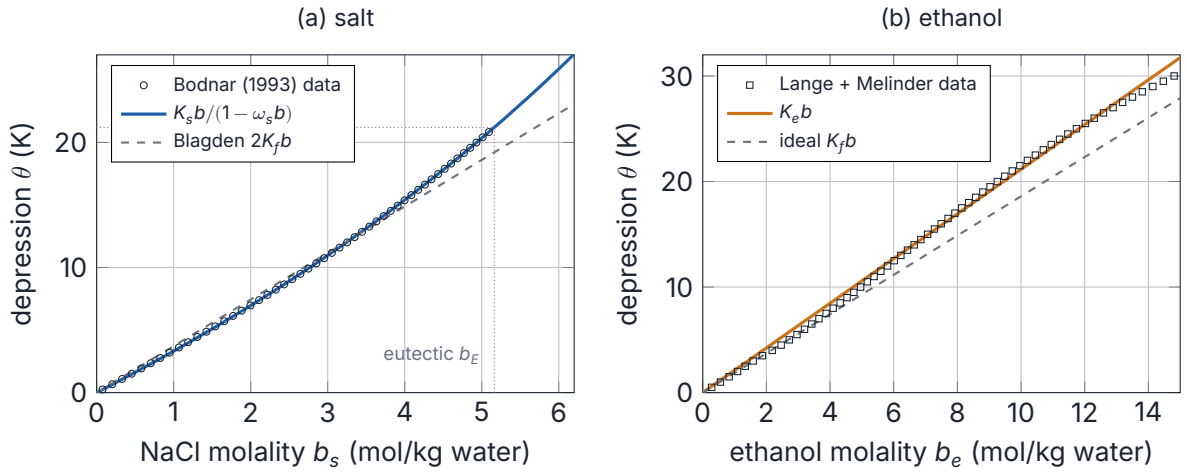


Figure 2. The two liquidus fits. (a) NaCl: the bound-water form follows Bodnar's data to 0.05 K; Blagden's law misses by up to 2.0 K. (b) Ethanol: a straight line $K_e b_e$ with an effective $K_e = 2.116$ (ideal: 1.86).

2.4 Regime U: salt all dissolved, ice left

All the active salt is in solution, $n_d = n'_s$, and $W = m_w + x$. Eq. (2.5) gives $W = W_b + N/\theta$. Substituting this into Eq. (2.2) and multiplying by $\theta\,(L_0 - \Delta c\,\theta)$ leaves a quadratic in θ:

$$A\,\theta^2 + B\,\theta - NL_0 = 0, \qquad \begin{aligned} A &= C_\theta - \Delta c\,(m_w - W_b), \\ B &= (m_w - W_b)\,L_0 + Q_0 + \Delta c\,N, \\ Q_0 &= C_r \Delta T_r - n'_s \Delta h_s - n'_e h_E + Q_x. \end{aligned} \tag{2.6}$$

$A > 0$ (it equals $m_w c_i + m_s c_s + m_e[\eta_e c_{ea} + (1-\eta_e)c_e] + m_i c_i + \Delta c\,W_b$) and $NL_0 > 0$, so exactly one root is positive:

$$\boxed{\theta_U = \frac{2NL_0}{B + \sqrt{B^2 + 4ANL_0}}}, \qquad T_b = -\theta_U, \qquad W = W_b + \frac{N}{\theta_U}, \qquad x = W - m_w. \tag{2.7}$$

(This form avoids the cancellation in $(-B + \sqrt{\cdot})/2A$.) It holds while two checks pass: ice is left ($x \le m_i$), and the brine is not saturated ($n'_s \le b_E\,(W - \lambda n'_e)$, see Eq. (2.9)). If $N = 0$ (plain ice water), $\theta = 0$ and $x = Q_0/L_0$.

2.5 Regime S: saturated, solid salt and ice both left

Salt only: the eutectic. Ice, solid salt and brine coexist: three phases of two components, so the phase rule gives $F = C - P + 1 = 2 - 3 + 1 = 0$ degrees of freedom. The temperature is pinned at $\theta = \theta_E$ and the brine at b_E, so $n_d = b_E W$. Eq. (2.2) becomes linear in W:

$$\boxed{W_S = \frac{m_w L_E + C_r \Delta T_r + C_\theta \theta_E + Q_x}{L_E + b_E \Delta h_s}}, \qquad L_E = L_0 - \Delta c\,\theta_E, \qquad m_{\text{salt,left}} = m_s - b_E W_S M_s. \tag{2.8}$$

With ethanol: the cotectic. Three components and three phases give $F = 1$, so the temperature now drifts with the ethanol content. Two facts fix it. Per kilogram of water, salt is less soluble when ethanol is present, $b_{\text{sat}} = b_E(1 - \lambda b_e)$, so $n_d = b_E(W - \lambda n'_e)$. Put that into Eq. (2.5):

$$\theta = \frac{PW + P'}{QW + Q'}, \qquad P = K_s b_E, \quad P' = n'_e(K_e - \lambda K_s b_E), \quad Q = 1 - \omega_s b_E, \quad Q' = n'_e(\lambda \omega_s b_E - \omega_e). \tag{2.9}$$

Multiplying Eq. (2.2) by $QW + Q'$ gives a quadratic $a_2 W^2 + a_1 W + a_0 = 0$:

$$\begin{aligned} a_2 &= \alpha_1 + b_E \Delta h_s Q, & \alpha_1 &= L_0 Q - \Delta c\,P, \\ a_1 &= \alpha_0 - m_w \alpha_1 - C_\theta P + b_E \Delta h_s Q' - GQ, & \alpha_0 &= L_0 Q' - \Delta c\,P', \\ a_0 &= -m_w \alpha_0 - C_\theta P' - GQ', & G &= C_r \Delta T_r + \lambda b_E n'_e \Delta h_s - n'_e h_E + Q_x. \end{aligned} \tag{2.10}$$

Since $a_2 > 0$ and $a_0 < 0$, there is one positive root, $W_S = (-a_1 + \sqrt{a_1^2 - 4a_2a_0})/2a_2$, and θ_S follows from Eq. (2.9). With $n'_e = 0$ this reduces exactly to Eq. (2.8). The cotectic constant $\lambda = 0.0461\,\mathrm{kg/mol}$ reproduces the NIST solubility data within 3.9 %. The cotectic temperature then agrees with the numerical model to 0.56 K.

2.6 Regime I: the ice is all gone

If the heat brought in is more than the ice can absorb, $x = m_i$ and $W = m_w + m_i$. Eq. (2.2) then gives the temperature directly:

$$\theta_I = \frac{m_i L_0 - C_r \Delta T_r + n_d \Delta h_s + n'_e h_E - Q_x}{C_\theta + m_i \Delta c} \qquad (\theta_I < 0 \text{ means above } 0\,^\circ\mathrm{C}). \tag{2.11}$$

A bath in regime I has no latent-heat store left, so it is useless for cooling a drink.

2.7 Which regime? Boundaries in closed form

Evaluate Eq. (2.7) first. If its two checks pass, that is the answer. If the brine comes out saturated, use regime S. If $x > m_i$, use regime I. Because the bath's enthalpy rises monotonically with the ice melted, exactly one regime is consistent. Setting the checks to equalities gives the boundaries as explicit formulas.

Salt needed to reach the eutectic (salt only). Put $\theta = \theta_E$ and $W = n'_s/b_E$ into Eq. (2.2). Both C_r and C_θ are linear in m_s, so

$$m_s^* = \frac{m_w\,[L_E + c_w(\Delta T_r + \theta_E)] + m_i c_i \theta_E}{\eta_s\,(L_E + b_E \Delta h_s)/(M_s b_E) - c_s(\Delta T_r + \theta_E)}. \tag{2.12}$$

For B1 ($m_w = 1.60$, $m_i = 4.0\,\mathrm{kg}$) this is 0.963 kg, so B1's 1.00 kg is just enough and leaves 36 g of solid salt. For your original charge B4 it is 1.409 kg: the 0.95 kg you planned never reaches the eutectic and the bath sits at −14.7 °C. This is the same "mass balance is not an energy balance" point as the 24 Oct note.

Ethanol that melts all the ice. Set $x = m_i$ in Eq. (2.2) and in Eq. (2.5) or Eq. (2.9). In the unsaturated branch $\theta = N/(m_w + m_i - W_b)$ is linear in m_e, so the energy balance becomes a quadratic in m_e; the saturated branch also reduces to a quadratic. For B1's water, salt and ice the ice is gone at $m_e = 1.46\,\mathrm{kg}$ ($T_b = -25.1\,^\circ\mathrm{C}$). For B2's at 1.55 kg (−29.5 °C), and for ethanol alone at 2.57 kg (−23.6 °C).

These are limits, not recipes

> At those points the bath has no ice left, and the temperatures are below −23.6 °C, where the NaCl liquidus is extrapolated (unreliable below −27 °C). The practical floors are the regime-S values with ice to spare.

2.8 How sensitive is the bath? Closed-form derivatives

Differentiate $F(\theta) = A\theta^2 + B\theta - NL_0 = 0$ implicitly. A, B and N are all linear in each mass, so for any input m_j

$$\frac{\partial \theta}{\partial m_j} = \frac{L_0\,\partial_j N - \theta^2\,\partial_j A - \theta\,\partial_j B}{2A\theta + B}, \tag{2.13}$$

with the partial derivatives

input	$\partial_j N$	$\partial_j A$	$\partial_j B$
m_w	0	c_i	$L_0 + c_w \Delta T_r$
m_i	0	c_i	0
m_s	$\eta_s K_s / M_s$	$c_s + \Delta c\,\eta_s \omega_s / M_s$	$c_s \Delta T_r + (\eta_s/M_s)(\Delta c\,K_s - \omega_s L_0 - \Delta h_s)$
m_e	$\eta_e K_e / M_e$	$\eta_e c_{ea} + (1-\eta_e) c_e$	$c_e \Delta T_r + (\eta_e/M_e)(\Delta c\,K_e - h_E)$

For your original charge B4 (regime U, −14.7 °C) the derivatives of the depression θ (positive means colder) are, per kilogram: water −5.00, ice −0.32, salt +14.70 and ethanol +5.91 K/kg. So 100 g more salt makes it about 1.5 K colder, and 100 g more warm water 0.5 K warmer.

The surprise: more ice does not make the bath colder

> $\partial\theta/\partial m_i = -c_i\theta^2/(2A\theta + B) < 0$ always. Ice at 0 °C has to be cooled to T_b like everything else, which melts more ice and dilutes the brine. Ice is the store that soaks up the *can's* heat, not the thing that sets T_b.

In regime S the temperature is pinned, so the inputs change how much ice melts instead. From Eq. (2.8), the liquid water rises by 1.563 kg per kg of water added, 0.142 per kg of ice and 0.129 per kg of salt.

2.9 Measuring the mixing coefficient: invert the closed form

In regime U, Eq. (2.5) with $W = m_w + x$ and x from Eq. (2.2) is *linear* in the active salt n'_s, for a known θ. So a thermometer reading of the settled bath gives η_s directly:

$$\eta_s = \frac{M_s}{m_s} \cdot \frac{\theta[m_w + (C_r\Delta T_r - n'_e h_E + C_\theta\theta)/(L_0 - \Delta c\,\theta) - \omega_e n'_e] - K_e n'_e}{K_s + \omega_s\theta + \theta\,\Delta h_s/(L_0 - \Delta c\,\theta)}. \tag{2.14}$$

Example. Fully mixed, B1 should settle at −21.2 °C. If your thermometer reads −18.0 °C, Eq. (2.14) gives $\eta_s = 0.80$: a fifth of the salt is not working. Measure this once per recipe and preparation method, then keep it.

2.10 Ethanol: one correction pass

The heat of mixing per mole and the apparent heat capacity of dissolved ethanol both change with concentration: $h_E = h_\infty + g\,b_e$ and $c_{ea} = \min(c_{\text{cap}},\, c_0 + c_1 b_e)$, with $b_e = n'_e/W$. Exact substitution would turn the quadratics into cubics. Instead, evaluate the closed form once with $b_e = 4.0$ mol/kg, compute $b_e = n'_e/W$ from that result, and evaluate once more. Without this second pass the bath is off by up to 5.0 K; with it, by at most 0.73 K.

2.11 Results, and the check against the numerical model

recipe	regime	T_b (°C)	x (kg)	ice left (kg)	solid salt (g)	b_s	b_e	ρ_b (kg/m^3)	numerical T_b
B0	U	0.00	0.502	2.50	–	–	–	1000	0.00
B1	S	−21.20	1.595	2.40	36	5.16	–	1177	−21.20
B2	S	−24.38	1.934	2.07	205	4.64	2.22	1142	−23.86
B3	U	−13.24	1.775	2.23	–	–	6.26	973	−13.21
B4	U	−14.68	1.629	1.37	0	3.84	–	1140	−14.67

The last column is the earlier numerical model (water-activity liquidus from the full data tables, temperature-dependent heat capacities and H^E, root-finding). Over 116 recipes (salt:ice 0–0.35, ethanol:ice 0–0.4, water:ice 0.2–0.65) the closed form is within 0.73 K (rms 0.27 K) and the ice left within 123 g. One borderline recipe is classed U instead of S (Fig. 4). The largest single difference among the five recipes is B2 (−0.51 K), from the cotectic.

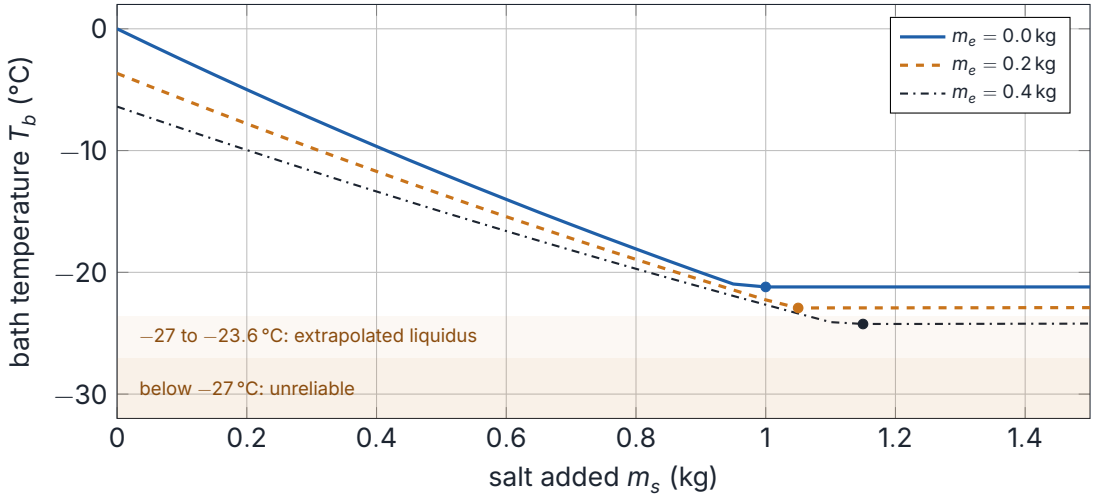


Figure 3. Bath temperature against salt for three ethanol loadings ($m_w = 1.6$, $m_i = 4.0$ kg). The dot marks where the brine saturates. With salt only, the curve then stays flat at the eutectic: it is a thermostat. With ethanol it keeps sinking slowly along the cotectic. Shading marks where the liquidus data run out.

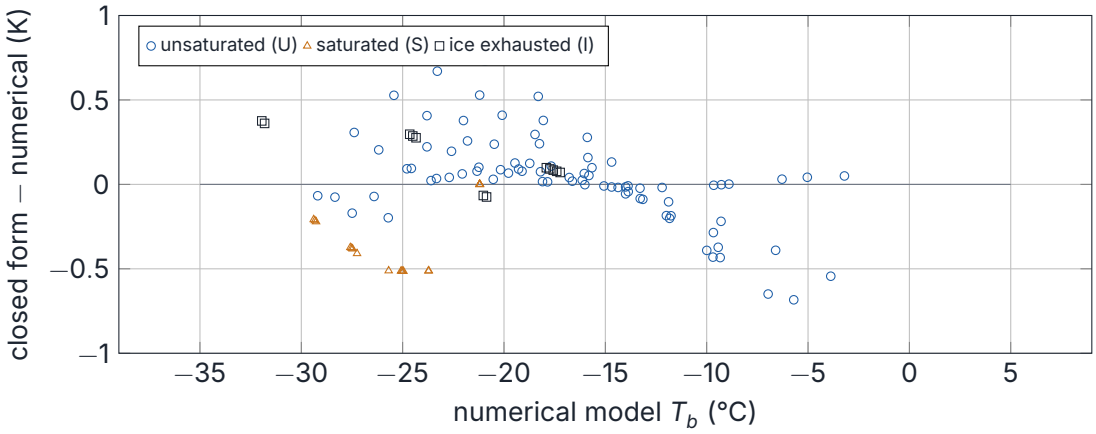


Figure 4. Closed form minus numerical model for 116 recipes, by regime.

3 Part II · The can, its air space, and where things sit in the tank

3.1 The part-filled can

Treat the can as a cylinder of outer radius R and height H, with a wall of thickness t_w. The drink of volume V_d stands at the bottom; above it is the air space (headspace gas).

$$r_i = R - t_w, \qquad h_d = \frac{V_d}{\pi r_i^2}, \qquad V_a = \pi r_i^2 H - V_d, \qquad h_a = H - h_d, \tag{3.1}$$

$$C = \rho_d V_d c_d + m_{\mathrm{Al}} c_{\mathrm{Al}}, \qquad M_c = \rho_d V_d + m_{\mathrm{Al}}, \qquad \bar{\rho}_c = \frac{M_c}{\pi R^2 H}. \tag{3.2}$$

h_d is the height of the drink column, V_a and h_a the volume and height of the air space, C the heat capacity that has to be cooled (drink plus aluminium), M_c the can's mass and $\bar{\rho}_c$ its mean density.

	Can 2 (standard)	Can 1 (slim)	
outer diameter, height	66.00, 116.6 mm	57.55, 146.2 mm	measured
drink column h_d (330 mL)	97.0 mm	127.7 mm	Eq. (3.1)
air space V_a (share of the inside), height h_a	66.5 mL (16.8 %), 19.6 mm	47.6 mL (12.6 %), 18.4 mm	Eq. (3.1)
heat capacity C	1 402 J/K	1 402 J/K	Eq. (3.2)
mass M_c, mean density $\bar{\rho}_c$	356 g, 893 kg/m^3	356 g, 937 kg/m^3	Eq. (3.2)
free-floating draft in eutectic brine / in water	88.5 / 104.1 mm	116.3 / 136.9 mm	Eq. (3.3)
force to hold it fully under eutectic brine	1.11 N	0.90 N	Eq. (3.3)

The air space does three separate things, and each enters the model in a different place:

1. it **lowers the heat capacity**: only the drink and the can need cooling (Eq. (3.2));
2. it **leaves a dry band of wall** at the top. Inside that band is gas, so the drink can only reach it by conduction along the aluminium. That band acts as a cooling fin (§3.4);
3. it **makes the can float**: $\bar{\rho}_c < \rho_b$, and here even $\bar{\rho}_c < 1000\,\mathrm{kg/m^3}$.

3.2 Buoyancy

A free can floats upright with draft d_f. To hold it at an immersion depth h_o you have to push it down with a force F:

$$d_f = \frac{M_c}{\rho_b\,\pi R^2}, \qquad F = \rho_b\, g\,\pi R^2 h_o - M_c g. \tag{3.3}$$

A 330 mL can floats even in fresh water: its draft there is 104.1 mm of its 116.6 mm. In eutectic brine its waterline sits at 88.5 mm, so the top 8.6 mm of the drink column would stick out, and it needs 1.11 N to hold it fully under. So it must be clamped, and the lid board in your rig note does exactly that. A floating can is also barely stable upright. Its centre of mass (near the middle of the drink) sits at about the height of its metacentre, so left alone it tips over.

3.3 Where the ice and the brine end up

From Part I take the ice left $m_l = m_i - x$, the brine mass $m_b = W + n_d M_s + n'_e M_e$, the brine density $\rho_b = \rho_0 + \beta_s w_s - \beta_e w_e$ (with w_s, w_e the salt and ethanol mass fractions), and any solid salt m_{ss}. Then

$$V_l = \frac{m_l}{\rho_i}, \quad V_b = \frac{m_b}{\rho_b} + \frac{m_{ss}}{\rho_{\mathrm{NaCl}}}, \quad A' = A_t - \pi R^2, \quad h_p = \frac{V_l}{\varphi_p A'}, \quad V_p = \frac{1-\varphi_p}{\varphi_p}\,V_l. \tag{3.4}$$

A' is the floor area beside the can, h_p the height the ice pack would occupy, and V_p the brine needed to fill all its pores. Ice is lighter than brine, so the layering depends on how you hold the ice (Fig. 1):

- **Ice floats freely.** By Archimedes a fraction ρ_i/ρ_b of the pack is under the surface, holding $V_f = (\rho_i/\rho_b)V_p$ of brine in its pores. The rest of the brine sits under the ice as a clear layer:

$$h_{cb} = \frac{V_b - V_f}{A'}, \qquad h_b = h_{cb} + \frac{\rho_i}{\rho_b}\,h_p, \qquad \text{ice above the surface: } \Big(1 - \frac{\rho_i}{\rho_b}\Big)h_p. \tag{3.5}$$

- **Ice pushed under a lid board.** The whole pack is submerged: $h_{cb} = (V_b - V_p)/A'$, and slush runs from h_{cb} up to $h_b = h_{cb} + h_p$.
- **Ice held down at the floor** (a mesh). Slush runs from the floor up to h_p, with clear brine above it up to $h_b = h_p + (V_b - V_p)/A'$.

If there is too little brine to float or submerge the pack ($V_b < V_f$, or $V_b < V_p$), the ice rests on the floor. Then $h_b = V_b/[(1-\varphi_p)A']$, slush runs from the floor to h_b, and dry ice sits above. Plain ice water B0 is like this. Finally, everything must fit: the top of the pack must stay below H_t.

recipe	ice held	clear brine h_{cb}	surface h_b	pack top	slush on can L_{sl}	brine on can L_{br}	fits?
B1	floats	5.0	14.8	17.6	5.7	4.0	at brim
B1	lid board	3.7	16.3	16.3	7.0	2.7	yes
B1	at floor	0	16.3	16.3	9.7	0.0	yes
B3	lid board	4.5	16.2	16.2	6.2	3.5	yes
B4	lid board	8.8	16.0	16.0	1.9	7.8	yes
B4	at floor	0	16.0	16.0	6.2	3.5	yes

All heights and lengths in cm, measured from the tank floor; the can's base is at $z_0 = 1.0$ cm.

B4 (your original charge) melts so much ice that with the lid board most of the drink column sits in clear brine: only 1.9 cm of the 9.7 cm drink column touches slush. That matters a lot in §4.

3.4 Contact zones on the can

The drink wets the wall from z_0 to $z_0 + h_d$. Overlapping that band with each layer gives the lengths of wall that have drink inside and slush, or clear brine, outside (L_{sl}, L_{br}). Wall facing ice above the brine surface, or air, is treated as insulated (A8). The bottom counts if the can is held off the floor ($z_0 > 0$).

$$A_{sl} = 2\pi R\, L_{sl}, \qquad A_{br} = 2\pi R\, L_{br}, \qquad A_{\mathrm{bot}} = \pi R^2. \tag{3.6}$$

The dry band as a fin. Above the drink line the wall has gas inside and liquid outside. It still cools the drink, because heat runs along the aluminium from the wetted wall. A thin fin of conductance $k_{\mathrm{Al}} t_w$ per unit width, losing heat to the outside with a mean coefficient $\bar{h}$, carries heat only over a length $1/m$:

$$m = \sqrt{\frac{\bar{h}}{k_{\mathrm{Al}} t_w}}, \qquad A_{\mathrm{fin}} = 2\pi R\, \frac{\tanh(mL_f)}{m}, \qquad A_{[x_a,x_b]} = 2\pi R\, \frac{\sinh m(L_f - x_a) - \sinh m(L_f - x_b)}{m \cosh mL_f}. \tag{3.7}$$

L_f is the length of dry wall in liquid. The last expression splits A_{fin} between layers when the band crosses from slush into brine. With $\bar{h} \approx 249\,\mathrm{W\,m^{-2}K^{-1}}$ (from §4), $1/m \approx 9$ mm. So only the first centimetre of the 19.6 mm dry band works, contributing $A_{\mathrm{fin}} \approx 18.2\,\mathrm{cm^2}$ of the 253.6 $\mathrm{cm^2}$ total contact area for B1 with the ice at the floor. $\bar{h}$ comes from the first pass of the cooling calculation, so the fin enters on the second pass.

4 Part III · Heat flow with no convection

The story in four steps

1. With nothing flowing, the salt next to the can cannot follow the meltwater. The brine there gets diluted, and the last ice beside the can melts at T_L, well above T_b (§4.2).
2. The slush does not melt at one temperature. Its latent heat is spread over a partly melted zone, so it behaves like a solid with a huge, temperature-dependent heat capacity (§4.3).
3. A classical similarity solution turns that zone into one number, an effective *effusivity* E (§4.4).
4. A well-mixed can in contact with such a material cools by an exact closed-form law, which comes out of a Laplace transform (§4.5).

4.1 What "no convection" does

Heat then moves only by conduction, $\partial H/\partial t = \nabla\cdot(k\nabla T)$ with H the enthalpy per unit volume, in the drink, the wall, the brine and the slush. The second consequence is easy to miss: **dissolved salt and ethanol stay put**. They could only move by diffusion, with $D \approx 1.5\times10^{-9}\,\mathrm{m^2/s}$, while heat diffuses with $\alpha \approx 1.3\times10^{-7}\,\mathrm{m^2/s}$. The ratio (the Lewis number) is about 90. In a 200 s cooling, salt moves about $\sqrt{Dt} \approx 0.5$ mm while heat reaches about 5 mm. So the water from ice melted by the can stays where it melted.

4.2 The diluted brine at the can: the coefficient you asked for

Take one cubic metre of slush next to the can, with ice volume fraction φ_p and its pores full of bath brine (mass fractions w_w, w_s, w_e):

$$W_0 = (1-\varphi_p)\rho_b w_w, \quad m_{I0} = \varphi_p \rho_i, \quad n_{s,v} = \frac{(1-\varphi_p)\rho_b w_s}{M_s}, \quad n_{e,v} = \frac{(1-\varphi_p)\rho_b w_e}{M_e}, \tag{4.1}$$

with $N_v = K_s n_{s,v} + K_e n_{e,v}$ and $W_{b,v} = \omega_s n_{s,v} + \omega_e n_{e,v}$. Because the solutes cannot leave, Eq. (2.5) applies to this cubic metre with its own fixed N_v. At depression θ it holds liquid water $W_l(\theta) = W_{b,v} + N_v/\theta$ and ice $m_l(\theta) = W_0 + m_{I0} - W_l(\theta)$. Far away it sits at θ_b with $W_l = W_0$. Heated, it melts progressively, and the last ice goes when $W_l = W_0 + m_{I0}$:

$$\theta_L = \frac{N_v}{W_0 + m_{I0} - W_{b,v}}, \qquad \boxed{\eta_c \equiv \frac{\theta_L}{\theta_b} = \frac{W_0 - W_{b,v}}{W_0 + m_{I0} - W_{b,v}}} \tag{4.2}$$

η_c is the ratio of free water in the brine alone to free water in brine plus ice. It is a "lowered coefficient" that needs no fitting: it follows from the no-convection assumption and the ice fraction. For B1 ($W_0 = 407$, $m_{I0} = 504$, $W_{b,v} = 92.5\,\mathrm{kg/m^3}$), $\eta_c = 0.384$. The last ice next to the can therefore melts at $T_L = -8.14\,°\mathrm{C}$, while the bath a few centimetres away is at $-21.2\,°\mathrm{C}$.

Two different coefficients, both lowered

> Your mixing coefficient η_s (§2.1) describes the *whole bath*: how much of the salt you poured actually dissolved. It sets T_b. The no-convection coefficient η_c describes the *few millimetres next to the can*: how much the can's own meltwater dilutes the brine there. They multiply: $\theta_L = \eta_c\,\theta_b(\eta_s)$. If undissolved salt crystals sit in the slush right against the can, η_c moves towards 1, because melting can then stay at the eutectic until the local crystals run out.

4.3 The slush is a heat sink with its latent heat spread out

Per unit volume, measuring from all-liquid at 0 °C, the enthalpy of the slush is

$$H(\theta) = -m_l(\theta)\,(L_0 - \Delta c\,\theta) - \theta\,C_v, \qquad C_v = c_w(W_0 + m_{I0}) + c_s M_s n_{s,v} + c_{ea} M_e n_{e,v}. \tag{4.3}$$

Its slope with temperature, the *apparent heat capacity*, is

$$c_{\mathrm{app}} = \frac{\mathrm{d}H}{\mathrm{d}T} = C_v - \Delta c\, m_l(\theta) + (L_0 - \Delta c\,\theta)\frac{N_v}{\theta^2} \qquad (\theta_L < \theta < \theta_b), \qquad c_{\mathrm{app}} = C_v \quad (\theta < \theta_L).$$

Three features matter.

- There is **no latent-heat jump** at T_L: the ice fraction falls continuously to zero there.
- Most of the latent heat is taken up near T_L, because N_v/θ^2 is largest there.
- The total heat needed to turn slush at T_b into liquid at T_L is

$$\Delta H = m_{I0}\,(L_0 - \Delta c\,\theta_b) + C_v\,(\theta_b - \theta_L) \qquad (\text{B1: } 196.7\,\mathrm{MJ/m^3},\ C_v = 3.92\,\mathrm{MJ\,m^{-3}K^{-1}}). \tag{4.4}$$

The numerical solution of the full conduction problem (Fig. 5) shows what this does. Behind the full-melt isotherm there is only a thin liquid layer: 3.1 mm after 240 s. Ahead of it a broad, partly melted zone reaches 1–2 cm, and it holds 37 % of the heat taken from the can after 60 s and 46 % after 240 s. A sharp-front (Stefan) model that puts all the latent heat at T_L ignores that zone, and comes out too slow: by +18 % at $\theta_b \approx 5\,\mathrm{K}$ and by +57 % at the eutectic (Appendix 6.2).

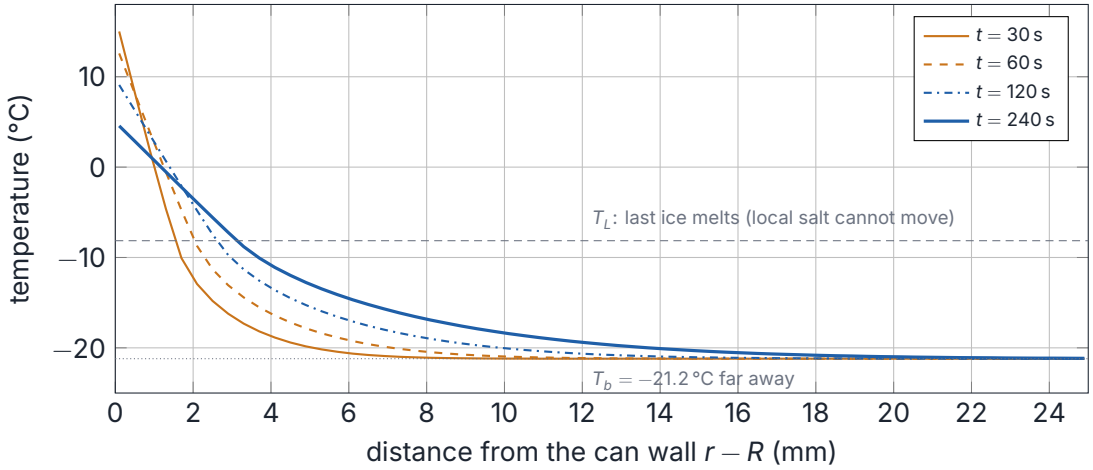


Figure 5. Temperature around a well-mixed Can 2 in B1 slush ($\varphi_p = 0.55$), from the numerical solution of the conduction problem. Left of the dashed line T_L the slush has fully melted (a few mm). To the right, a broad partly melted zone at temperatures between T_L and T_b soaks up nearly half of the heat.

4.4 One number for the slush: its effective effusivity

The classical problem. Hold a flat wall at temperature T_w against a half-space of slush at T_b. With no length scale in the problem, the solution depends only on $\xi/\sqrt{t}$ (ξ is the distance from the wall). Treat the liquid layer (conductivity k_l, heat capacity C_v, diffusivity $\alpha_l = k_l/C_v$) and the partly melted zone separately. Give the second constant effective properties: $c_m = \Delta H/(\theta_b - \theta_L)$, and k_m = the mean slush conductivity between T_b and T_L (ice fraction weighted, Maxwell–Eucken), with $\alpha_m = k_m/c_m$. The isotherm T_L then moves as $s = 2\lambda\sqrt{\alpha_l t}$ and

$$T = T_w - (T_w - T_L)\frac{\operatorname{erf}\left(\xi/2\sqrt{\alpha_l t}\right)}{\operatorname{erf}\lambda}\ (\xi < s), \qquad T = T_b + (T_L - T_b)\frac{\operatorname{erfc}\left(\xi/2\sqrt{\alpha_m t}\right)}{\operatorname{erfc}\left(\lambda\sqrt{\alpha_l/\alpha_m}\right)}\ (\xi > s).$$

There is no latent jump at T_L, so the heat flux must be continuous there. That fixes λ:

$$\frac{E_l\,(T_w - T_L)\,e^{-\lambda^2}}{\operatorname{erf}\lambda} = \frac{E_m\,(T_L - T_b)\,e^{-\lambda^2\alpha_l/\alpha_m}}{\operatorname{erfc}(\lambda\sqrt{\alpha_l/\alpha_m})}, \qquad E_l = \sqrt{k_l C_v}, \quad E_m = \sqrt{k_m c_m}. \tag{4.5}$$

This is one equation in one unknown, solved once (B1: $\lambda = 0.296$). The flux through the wall is then

$$q = \frac{E_l(T_w - T_L)}{\operatorname{erf}\lambda\,\sqrt{\pi t}} \equiv \frac{E\,(T_w - T_b)}{\sqrt{\pi t}}, \qquad \boxed{E = \frac{E_l\,(T_w - T_L)}{(T_w - T_b)\,\operatorname{erf}\lambda}} \tag{4.6}$$

E is exactly what a uniform material of effusivity E would give. So, as far as the wall can tell, the slush behaves like such a material, with an effective conductivity and heat capacity

$$k = \frac{E^2}{c_{\text{sec}}}, \qquad c_{\text{sec}} = \frac{\Delta H + C_v(T_w - T_L)}{T_w - T_b}. \tag{4.7}$$

For B1: $E_l = 1\,428$, $E_m = 3\,685$ and $E = 2\,576\,\mathrm{W\,s^{1/2}\,m^{-2}\,K^{-1}}$, with $k = 0.78\,\mathrm{W\,m^{-1}K^{-1}}$. In plain ice water ($\theta_b = 0$) the ice melts at one temperature, so Eq. (4.5) becomes the textbook one-phase Stefan condition $\lambda e^{\lambda^2}\operatorname{erf}\lambda = \mathrm{Ste}/\sqrt{\pi}$, with $\mathrm{Ste} = C_v T_w/(m_{l0}L_0)$.

Which wall temperature? The drink cools, so T_w falls from T_0 to T_1 and E changes along the way. Two limits fix the choice. When the partly melted range $\theta_b - \theta_L$ is wide compared with how far the drink ends above T_L, the slush behaves almost linearly and the time-averaged drink temperature $\bar{T}$ is right. When that range is narrow (fresh water), the slush behaves like a sharp front, the last stage dominates, and T_1 is right. Interpolating between them,

$$T_{\text{ref}} = w\,\bar{T} + (1 - w)\,T_1, \qquad w = \min(1, \psi), \qquad \psi = \frac{\theta_b - \theta_L}{T_1 - T_L}, \tag{4.8}$$

with $\bar{T}$ from Eq. (4.12) below (two or three passes converge). B1 has $\psi = 0.99$, so $T_{\text{ref}} = 10.3\,°\text{C}$. Plain ice water has $\psi = 0$, so $T_{\text{ref}} = T_1$. This rule is the only empirical step in Part III. Without it, the error against the full solution rises from 10.7 % to about 50 % for nearly fresh baths.

4.5 The cooling law

Set-up. Let $\Theta(t) = T_d(t) - T_b$ be the drink's excess over the bath. A well-mixed drink loses heat through its contact area A:

$$C\,\frac{\mathrm{d}\Theta}{\mathrm{d}t} = -A\,q(t).$$

For a medium of effusivity E and conductivity k around a cylinder of radius R, the response to a wall excess $\Theta_w(t)$ is, in Laplace variables ($\hat{f}(s) = \int_0^\infty f e^{-st}\mathrm{d}t$),

$$\hat{q}(s) = \Big(E\sqrt{s} + \frac{k}{2R}\Big)\hat{\Theta}_w(s).$$

The first term is the flat-wall response, $q = E\Theta_w/\sqrt{\pi t}$ for a step. The second is the first correction for the curved surface (for a step it adds the steady $k\Theta_w/2R$). The aluminium has no resistance and the drink is well mixed, so $\Theta_w = \Theta$. Transforming the drink equation, with $\Theta(0) = \Theta_0$:

$$C(s\hat{\Theta} - \Theta_0) = -A\Big(E\sqrt{s} + \frac{k}{2R}\Big)\hat{\Theta} \quad\Longrightarrow\quad \hat{\Theta} = \frac{\Theta_0}{s + a\sqrt{s} + b} = \frac{\Theta_0}{(\sqrt{s}+p_1)(\sqrt{s}+p_2)},$$

$$a = \frac{AE}{C}, \qquad b = \frac{A_{\text{lat}}\,k}{2RC}, \qquad p_{1,2} = \frac{a \mp \sqrt{a^2 - 4b}}{2}. \tag{4.9}$$

Split into partial fractions, $1/[(\sqrt{s}+p_1)(\sqrt{s}+p_2)] = [1/(\sqrt{s}+p_1) - 1/(\sqrt{s}+p_2)]/(p_2 - p_1)$. Then use the standard pair $\mathcal{L}^{-1}\{1/(\sqrt{s}+p)\} = 1/\sqrt{\pi t} - p\,\operatorname{erfcx}(p\sqrt{t})$, where $\operatorname{erfcx}(z) = e^{z^2}\operatorname{erfc}(z)$. The $1/\sqrt{\pi t}$ terms cancel, leaving

$$\boxed{T_d(t) = T_b + \Theta_0\,\frac{p_2\,\operatorname{erfcx}(p_2\sqrt{t}) - p_1\,\operatorname{erfcx}(p_1\sqrt{t})}{p_2 - p_1}} \tag{4.10}$$

Differentiating with $\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{erfcx}(p\sqrt{t}) = p^2\operatorname{erfcx}(p\sqrt{t}) - p/\sqrt{\pi t}$ gives the cooling rate. The time average needed in Eq. (4.8) also has a closed form:

$$-\frac{\mathrm{d}T_d}{\mathrm{d}t} = \Theta_0\Big[\frac{a}{\sqrt{\pi t}} - \frac{p_2^3\operatorname{erfcx}(p_2\sqrt{t}) - p_1^3\operatorname{erfcx}(p_1\sqrt{t})}{p_2 - p_1}\Big], \tag{4.11}$$

$$\bar{\Theta} = \frac{1}{t_1}\int_0^{t_1}\Theta\,\mathrm{d}t = \frac{\Theta_0}{t_1(p_2 - p_1)}\Big[\frac{\operatorname{erfcx}(p_2\sqrt{t_1}) - 1}{p_2} - \frac{\operatorname{erfcx}(p_1\sqrt{t_1}) - 1}{p_1}\Big], \qquad \bar{T} = T_b + \bar{\Theta}. \tag{4.12}$$

The time to reach the target, t_5, is the single root of $T_d(t_5) = T_1$ in Eq. (4.10), which falls monotonically. The minimum cooling rate inside the window $[T_0, T_1]$ is Eq. (4.11) at t_5.

What the law says.

- **Short times:** $\Theta \approx \Theta_0(1 - 2a\sqrt{t/\pi})$. The rate starts infinite and falls like $1/\sqrt{t}$, because a cold medium pressed against a hot wall has no boundary layer yet. In practice the first second is limited by contact and by how well the drink mixes.
- **Long times:** $\Theta \approx \Theta_0\,a/(2\sqrt{\pi}\,b^2 t^{3/2})$, a power law rather than an exponential.
- **The effective conductance falls with time:** $q/\Theta \approx E/\sqrt{\pi t} + k/2R$, about $200\,\text{W}\,\text{m}^{-2}\text{K}^{-1}$ after one minute in B1 and about half that after four. This is why "a time constant τ" does not exist here.
- **Real or complex roots:** for a can wholly in one medium, $C \approx \pi R^2 h\rho_d c_d$ and $A_{\text{lat}} = 2\pi Rh$. Then $a^2/4b = c_{\text{sec}}/(\rho_d c_d)$, so the roots are real when the medium stores more heat per kelvin than the drink. Slush does ($c_{\text{sec}} \approx 2\rho_d c_d$). Clear brine is marginal ($a^2/4b = 0.96$): the roots are complex conjugates, and Eq. (4.10) still holds with the complex erfcx (the Faddeeva function, `scipy.special.wofz`).

4.6 Several contact zones, the bottom and the fin

Each zone of wall sees the same drink temperature, and the responses add. So the closed form keeps its shape, with

$$a = \frac{(A_{sl} + A_{\mathrm{fin},sl})E + (A_{br} + A_{\mathrm{fin},br})E_{br} + A_{\mathrm{bot}}E_{\mathrm{bot}}}{C}, \qquad b = \frac{(A_{sl} + A_{\mathrm{fin},sl})k + (A_{br} + A_{\mathrm{fin},br})k_b}{2RC}, \tag{4.13}$$

where the clear brine is an ordinary conductor, $E_{br} = \sqrt{k_b \rho_b c_b}$ (c_b is the brine's specific heat; $E_{br} = 1\,447$ for B1), and the flat bottom contributes no curvature term. The fin needs $\bar{h}$, which comes from the first pass:

$$\bar{h} = \frac{C\,(\Theta_0 - \Theta_1)}{A\,t_5\,\bar{\Theta}}. \tag{4.14}$$

Evaluation order: first $T_{\mathrm{ref}} \to E, k \to a, b \to t_5, \bar{T}, \bar{h}$; then T_{ref} and A_{fin} again; repeat once more.

4.7 Is the closed form right? Check against the full conduction solution

I solved the full problem numerically with no closed-form steps, as an independent check. It is a one-dimensional radial enthalpy method: the drink (well mixed, or conducting), the aluminium, and slush with immobile solutes, using its exact $H(T)$ from Eq. (4.3) and a Maxwell–Eucken conductivity. Halving the grid spacing and quartering the time step changes the answer by less than 0.1 s.

case	θ_b (K)	φ_p	ψ	full solution t_5 (s)	closed form t_5 (s)	error (%)
plain ice water	0.00	0.55	0.00	908	959	+5.7
weak salt 50 g	1.28	0.55	0.12	774	798	+3.1
weak salt 100 g	2.54	0.55	0.22	681	695	+2.1
salt 200 g	4.99	0.55	0.38	553	558	+0.9
salt 300 g	7.37	0.55	0.50	468	468	−0.1
salt 500 g	11.87	0.55	0.69	361	355	−1.7
salt 700 g	16.08	0.55	0.84	295	286	−3.0
B1 eutectic	21.20	0.55	0.99	239	229	−4.4
B1, loose slush	21.20	0.35	0.50	310	303	−2.3
B1, dense slush	21.20	0.65	1.35	216	192	−10.7
B2 salt+ethanol	24.38	0.55	1.09	211	199	−5.9
B3 ethanol only	13.24	0.55	0.77	336	329	−2.2
B4 rig charge	14.68	0.55	0.79	315	306	−2.6
B1, Can 1	21.20	0.55	0.99	182	174	−4.3
B4, Can 1	14.68	0.55	0.79	239	233	−2.6
plain ice water, Can 1	0.00	0.55	0.00	689	729	+5.8

Over 16 cases, from plain ice water to the cotectic, two can sizes and three ice fractions, the closed form for the lateral surface has mean error −1.4 %, rms 4.4 % and worst 10.7 % (dense slush, $\varphi_p = 0.65$). For clear brine alone it is −1.1 % (638 vs 645 s). Dropping the curvature term would be +29 % off.

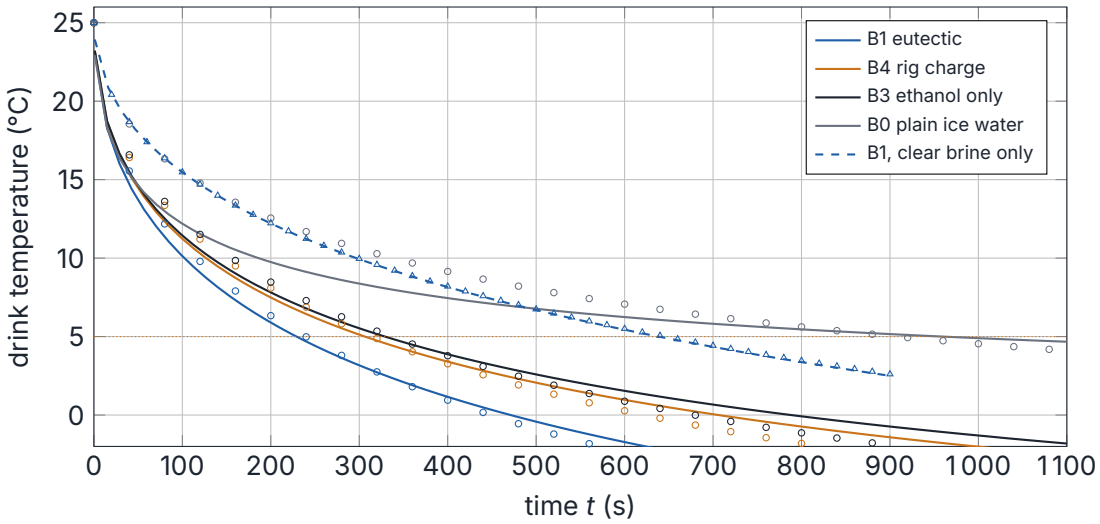


Figure 6. Drink temperature for a well-mixed Can 2 (side wall only, as in the check), from Eq. (4.10) (lines) and from the full numerical solution (symbols).

4.8 Results for your rig

Full can (side wall, bottom and fin), Can 2 with 330 mL, base 1.0 cm off the floor, $\varphi_p = 0.55$, cooling $25 \to 5$ °C:

recipe	ice held	T_b (°C)	E slush ($\mathrm{W\,s^{1/2}m^{-2}K^{-1}}$)	t_5 (s)	rate at 15 °C (K/min)	rate at 5 °C (K/min)	mean rate (K/min)
B0 plain ice	floor	0.0	5 446	640	6.9	0.22	1.87
B1 eutectic	floats	−21.2	2 574	223	7.0	1.79	5.38
	board		2 575	203	7.7	1.95	5.91
	floor		2 576	151	10.4	2.59	7.93
B2 salt + ethanol	board	−24.4	2 496	185	8.2	2.23	6.48
	floor		2 497	138	11.1	2.97	8.73
B3 ethanol only	board	−13.2	2 960	324	5.4	1.06	3.71
	floor		2 963	222	8.1	1.52	5.41
B4 original charge	board	−14.7	2 867	447	3.8	0.81	2.68
	floor		2 870	287	6.0	1.23	4.18

What the table says:

- **Salt pays**, even with no convection. Going from plain ice water to the eutectic cuts the time by 4.2 × with the ice at the floor.
- **Ice placement pays almost as much.** Slush around the whole drink column beats a floating pack by 1.47 ×. Clear brine has no latent store, so its effusivity is only 1 447.
- **Ethanol does not pay here.** B2 is only slightly faster than B1 despite being 3 K colder. B3 (ethanol only) is much slower because it is warmer and has melted more of the ice.
- **B4 is slow for a geometric reason as much as a thermal one:** with a lid board almost all of its drink column sits in clear brine.

A check against published data. SINTEF timed a 0.33 L can from 21 to 5 °C: about 600 s in ice water and 150 s in a −16 °C salt bath. The same calculation (21 °C start, Can 2, ice at the floor, $m_s = 0.696$ kg for −16 °C) gives 445 s and 128 s (154 s with a lid board). With no convection inside the drink either, the full numerical solution gives 2 197 s and 752 s. Both measurements sit near the well-mixed limit and far from the no-convection-inside limit: a real drink mixes itself well by natural convection. The ice-water measurement is about a third slower than the well-mixed prediction, which is what you would expect if water rather than packed slush surrounded part of the can.

4.9 No convection inside the drink either

Taken literally, “disregard convection” also stops the drink from mixing, so heat has to diffuse through it too. This is the **true minimum cooling rate**: any convection, inside or outside, can only speed things up.

The contact temperature. Two materials of effusivity $E_d = \sqrt{k_d \rho_d c_d}$ (drink, 1536) and E (slush) brought into contact at once jump to a common surface temperature. To leading order it stays there:

$$\boxed{T_c = \frac{E_d\, T_0 + E\, T_b}{E_d + E}} \tag{4.15}$$

For B1, $T_c = -7.38\,°\mathrm{C}$; the full numerical solution’s wall temperature after 300 s is $-7.37\,°\mathrm{C}$.

The drink at the wall must supercool or freeze

> T_c is far below the drink’s freezing point ($-0.77\,°\mathrm{C}$). In a no-convection world the layer of drink against the wall either supercools, which canned drinks often do, or freezes onto the wall while the centre is still near 25 °C. The numbers below assume it supercools; freezing would add a second moving front.

Closed form for the mean drink temperature. Inside a solid cylinder the response, to the same order, is $E_d\sqrt{s} - k_d/2r_i$. The curvature now *reduces* the flux, because the inside runs out of heat. Matching it to the outside response $E\sqrt{s} + k/2R$ at the wall and inverting the transform (Appendix 6.1) gives

$$\boxed{\frac{\bar{\Theta}(t)}{\Theta_0} = 1 - \frac{A}{C\,D}\left[E_d E f_2 + \beta f_3 - \gamma f_4\right]} \tag{4.16}$$

$$D = E_d + E, \qquad g_d = \frac{k_d}{2r_i}, \qquad g_s = \frac{k}{2R}, \qquad c = \frac{g_s - g_d}{D}, \qquad \beta = E_d g_s - E g_d, \qquad \gamma = g_d g_s,$$

$$f_1 = \operatorname{erfcx}(c\sqrt{t}), \qquad f_2 = \frac{1 - f_1}{c}, \qquad f_3 = \frac{1}{c}\left(2\sqrt{\frac{t}{\pi}} - f_2\right), \qquad f_4 = \frac{1}{c}\,(t - f_3).$$

It needs a single erfcx and holds while the drink’s Fourier number $\alpha_d t / r_i^2 \lesssim 0.15$. For a linear outside medium it agrees with the full solution to 4 %. In slush it is −13 % (B1) and −9 % (B4) against the full solution, because the wall temperature drifts down through the partly melted range. Use the full-solution numbers as the bound:

	T_c (°C)	t_5 mean, full solution	t_5 mean, Eq. (4.16)	t_5 centre, full solution	rate at 5 °C (full)
B1	−7.38	813 s	711 s	1940 s	0.66 K/min
B4	−3.57	1052 s	954 s	2193 s	0.49 K/min

So for B1 the drink’s mean temperature takes about 813 s, and its centre 1940 s, compared with 151–223 s for a well-mixed drink. These are side-wall-only numbers, which is conservative, as a bound should be.

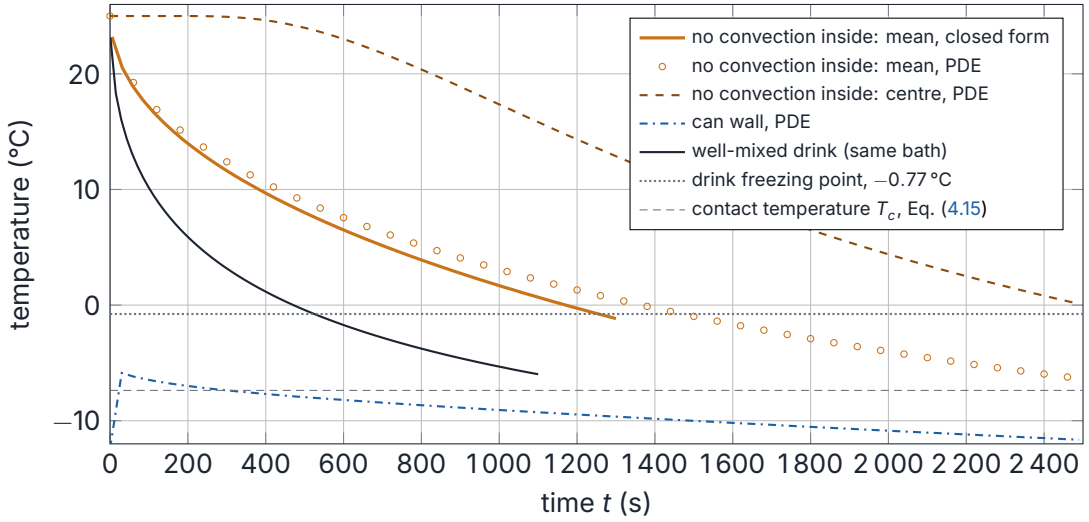


Figure 7. B1, Can 2, side wall only: with no convection inside the drink, the wall drops at once to about T_c and stays below the drink's freezing point. The mean temperature falls like $\sqrt{t}$ and the centre lags far behind. The well-mixed drink in the same bath is shown for comparison.

5 The whole chain, design and what to measure

5.1 From your four masses to the drink temperature

Evaluation order: each line uses only quantities from the lines above and the measured constants

1.	Active solute n'_s, n'_e	Eq. (2.1)
2.	Heat capacities C_r, C_θ	Eq. (2.3)
3.	Depression strength N and bound water W_b	Eq. (2.5)
4.	A, B, Q_0; bath depression θ_U; then W, x	Eqs. (2.6), (2.7)
5.	If saturated: W_S and θ_S. If $x > m_i$: θ_l	Eqs. (2.8)–(2.11)
6.	With ethanol: update h_E, c_{ea} at $b_e = n'_e/W$ and repeat 1–5 once	§2.10
7.	Brine: $b_s = n_d/W$, $b_e = n'_e/W$, m_b, w_s, w_e, ρ_b	§3.3
8.	Can: r_i, h_d, V_a, C, M_c	Eqs. (3.1), (3.2)
9.	Tank: V_l, V_b, h_p, V_p, then the layers for your way of holding the ice	Eqs. (3.4), (3.5)
10.	Contact lengths and areas A_{sl}, A_{br}, A_{bot}	Eq. (3.6)
11.	Slush at the can: W_0, m_{l0}, N_v, $W_{b,v}$; then θ_L, η_c, C_v, ΔH, c_m, k_m	Eqs. (4.1)–(4.4)
12.	Effusivity: T_{ref}, then λ, E, k; clear brine E_{br}	Eqs. (4.8), (4.5)–(4.7)
13.	Cooling: a, b, $p_{1,2}$; $T_d(t)$; t_5; rate; $\bar{T}$; $\bar{h}$; fin A_{fin}	Eqs. (4.9)–(4.14), (3.7)
14.	Repeat 12–13 twice (the reference temperature and the fin settle)	
15.	Strict lower bound, no convection inside: T_c and $\bar{\Theta}(t)$	Eqs. (4.15), (4.16)

The only numerical steps are two scalar roots, λ in Eq. (4.5) and t_5 in Eq. (4.10); both functions are monotonic. Everything else is explicit.

5.2 What depends on what

derived quantity	is a function of	equation
D n_s', n_e'	m_s, m_e; η_s, η_e	(2.1)
D C_r, C_θ	m_w, m_s, m_e, m_i; c's, η_e	(2.3)
D N, W_b	n_s', n_e'; $K_s, K_e, \omega_s, \omega_e$	(2.5)
D θ_b $(= -T_b)$	m_w, m_s, m_e, m_i; T_r; $L_0, \Delta c, \Delta h_s, h_E, K$'s, ω's, b_E, θ_E, λ; η_s, η_e	(2.7), (2.8), (2.9), (2.11)
D W, x, n_d	as θ_b	same
D ρ_b, m_b, V_b, V_l	W, n_d, n_e', x; $\rho_0, \beta_s, \beta_e, \rho_i$	(3.4)
D h_d, V_a, C, M_c	$V_d, R, H, t_w, m_{\mathrm{Al}}$; $\rho_d, c_d, c_{\mathrm{Al}}$	(3.1), (3.2)
D layers h_{cb}, h_b, h_p	V_b, V_l, ρ_b; A_t, R, φ_p; how the ice is held	(3.5)
D L_{sl}, L_{br}, A's	layers; z_0, h_d, H, R	(3.6)
D θ_L, η_c	θ_b, brine composition, ρ_b; φ_p, ρ_i	(4.2)
D ΔH, C_v, c_m, k_m	θ_b, θ_L, slush composition; $L_0, \Delta c$, c's, k_i, k_b	(4.3), (4.4)
D λ, E, k	$T_{\mathrm{ref}}, T_b, T_L$; k_l, C_v, k_m, c_m	(4.5)–(4.7)
D a, b, $p_{1,2}$	A's, E, k, E_{br}, k_b, C, R	(4.9), (4.13)
D $T_d(t)$, t_5, rate	a, b, T_b, T_0; T_1	(4.10), (4.11)
D T_c, $\bar{\Theta}(t)$ (no convection inside)	$E_d, E, k, k_d, r_i, R, A/C, T_0, T_b$	(4.15), (4.16)

5.3 Design: what each mass does to the time

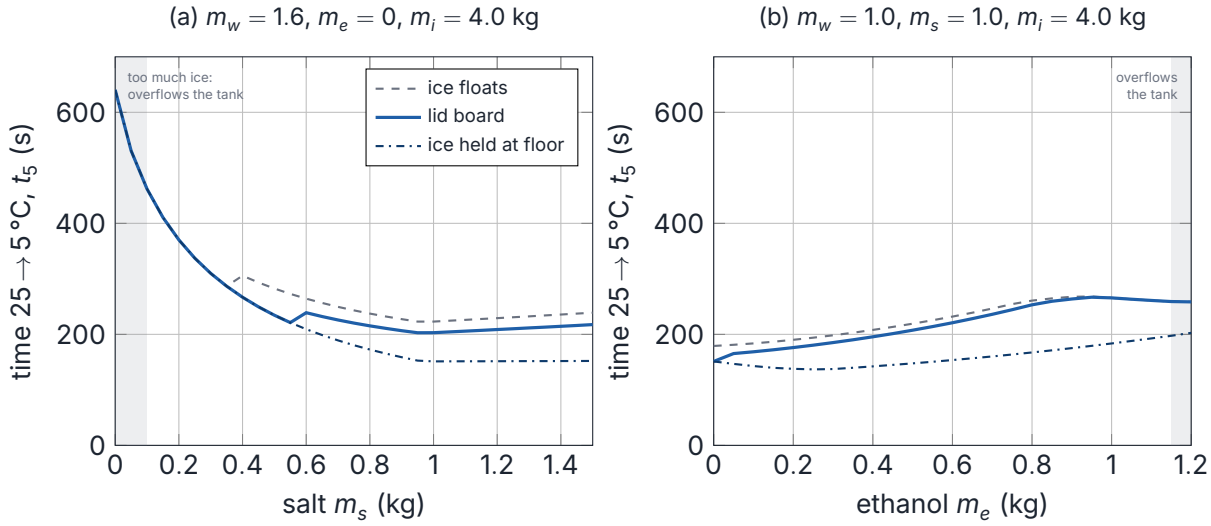


Figure 8. Time to cool Can 2 from 25 to 5 °C against (a) salt and (b) ethanol, for the three ways of holding the ice. Grey bands: charges that do not fit your 17.5 cm tank with the can in. The small steps are where clear brine first reaches the can's base, which then sits in brine instead of slush.

- **Salt (a).** The time falls steeply until the brine saturates, near m_s^* (Eq. (2.12)). Past that point the bath is a thermostat, and extra salt only brings in warm mass that melts a little more ice. With the ice floating or under a board, that ice comes off the pack around the can.
- **Ethanol (b).** Each 100 g lowers T_b, but each 100 g also melts ice through its warmth and its heat of mixing. With no convection the second effect wins sooner than you would guess, because the can ends up in clear brine.
- **Ice.** Below saturation, more ice at 0 °C makes the bath very slightly *warmer* (§2.8). What it really changes is how much slush there is around the can.
- **Water.** In regime U, each kilogram of warm water raises T_b by 5.0 K. In regime S the temperature stays put, and each kilogram of water instead adds 1.563 kg of liquid water (itself plus the ice it melts).

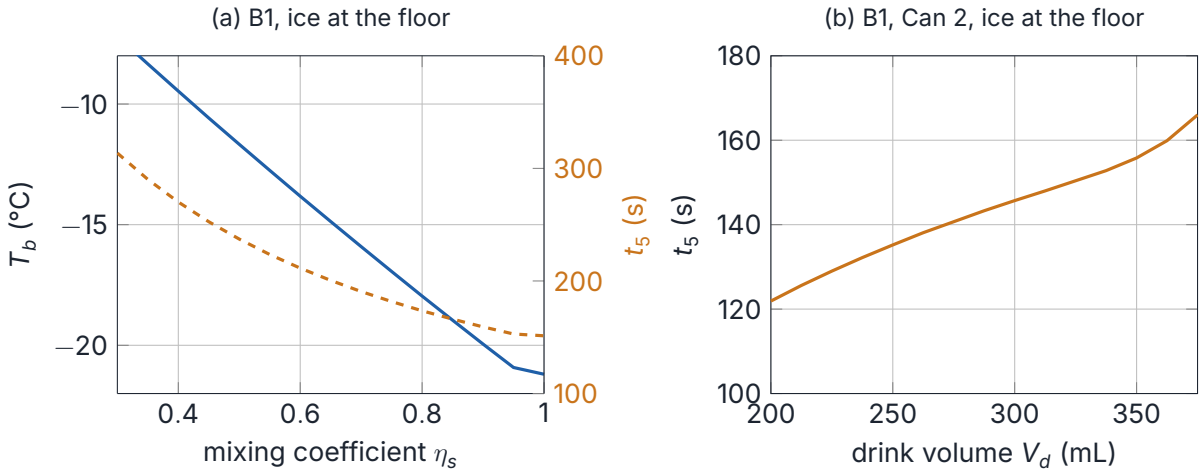


Figure 9. (a) B1 with the ice at the floor: bath temperature (blue, left axis) and time to 5 °C (amber dashed, right axis) against the salt mixing coefficient η_s. (b) Time to 5 °C against the volume of drink in Can 2: nearly doubling the drink changes the time by about a third.

With $\eta_s = 0.3$ (badly mixed salt) B1 only reaches −7.2 °C and needs 314 s. Changing the fill from 200 to 375 mL moves t_5 only from 122 to 166 s, because the drink's heat capacity and its wetted wall both scale with the fill. What remains comes from the fixed bottom, the aluminium and the shrinking fin.

5.4 What to measure, and how

1. φ_p**, the ice packing.** Fill a jug with crushed ice to a mark, top it up with ice water to the same mark, and record the water added: $\varphi_p = 1 - V_{\text{water}}/V_{\text{mark}}$. It enters η_c, the layering and E.
2. η_s**, the salt mixing.** Let the bath settle (no stirring) and read T_b at least 5 cm from the can, then apply Eq. (2.14). Do it for each way of adding the salt (all at once, in layers, pre-dissolved).
3. **The layering.** Mark h_{cb} and h_b on the tank wall and compare with Eq. (3.5). That tests φ_p and ρ_b in one go.
4. T_L**, the diluted brine.** A fine thermocouple bead 2–3 mm from the can wall should read between T_L and T_b while the can cools (Fig. 5). If it reads close to T_b, convection is carrying the meltwater away and η_c is closer to 1 than Eq. (4.2) says.
5. **The shape of the cooling curve.** Fit Eq. (4.10) with E as the only free parameter, and also fit an exponential. The ratio $E_{\text{fit}}/E_{\text{predicted}}$ is a direct measure of how much convection actually helps in your tank: 1 means none.

5.5 Limits of this model, bluntly

- **Real baths convect.** Meltwater is fresh and light and rises; brine is heavy and sinks. In reality the diluted layer is partly swept away, so the local coefficient lies between η_c and 1. Treat this model as the no-flow limit, not a forecast of a stirred tank.
- **The well-mixed drink** is an assumption. A real drink sits between the two limits given here, near the well-mixed one according to the SINTEF data.
- **The slush as a linear medium** is an approximation: within ±10.7 % of the full conduction solution in 16 cases. The reference-temperature rule, Eq. (4.8), is the one empirical step. The bottom and the fin are added by superposition and were not checked against a 2-D solution.
- **Bath thermodynamics.** The liquidus is good to 1.31 K, and the cotectic below −23.6 °C is extrapolated (B2). Δh_s, c_s and the ethanol mixing table are placeholders carried over from the earlier model; the H^E table came from a scrambled extraction and needs checking against the paper. Hydrohalite formation and dissolution kinetics are not modelled; η_s absorbs them empirically.
- **Geometry.** The can is a cylinder (dome and neck ignored) in perfect contact with the slush. The ice pack is uniform, although real packs are looser near walls. The tank height, 17.5 cm, was inferred,

not measured.

6 Appendix

6.1 Laplace pairs and the two inversions

With $\hat{f}(s) = \int_0^\infty f(t)e^{-st}\,\mathrm{d}t$, the pairs used are

$$\mathcal{L}\{1\} = \frac{1}{s}, \quad \mathcal{L}\{t\} = \frac{1}{s^2}, \quad \mathcal{L}\Big\{\frac{1}{\sqrt{\pi t}}\Big\} = \frac{1}{\sqrt{s}}, \quad \mathcal{L}\Big\{2\sqrt{\tfrac{t}{\pi}}\Big\} = \frac{1}{s^{3/2}}, \quad \mathcal{L}\{\mathrm{erfcx}(p\sqrt{t})\} = \frac{1}{\sqrt{s}\,(\sqrt{s}+p)}.$$

Writing $1/(\sqrt{s}+p) = 1/\sqrt{s} - p/[\sqrt{s}(\sqrt{s}+p)]$ gives $\mathcal{L}^{-1}\{1/(\sqrt{s}+p)\} = 1/\sqrt{\pi t} - p\ \mathrm{erfcx}(p\sqrt{t})$, the pair used for Eq. (4.10).

Well-mixed can (§4.5). $\hat{\Theta}/\Theta_0 = 1/[(\sqrt{s}+p_1)(\sqrt{s}+p_2)]$. Partial fractions and the pair above give Eq. (4.10), valid for real or complex-conjugate $p_{1,2}$.

Drink with no convection inside (§4.9). Let $\sigma = \sqrt{s}$. The drink, initially uniform at excess Θ_0, delivers $\hat{q} = Y_d(\Theta_0/s - \hat{\Theta}_w)$ to the wall, with $Y_d = E_d\sigma - g_d$. The outside takes $\hat{q} = Y_s\hat{\Theta}_w$, with $Y_s = E\sigma + g_s$. Equating the two:

$$\hat{\Theta}_w = \frac{Y_d\Theta_0}{s(Y_d+Y_s)}, \qquad \hat{q} = \frac{Y_dY_s\,\Theta_0}{s\,(Y_d+Y_s)}, \qquad \frac{\hat{\bar{\Theta}}}{\Theta_0} = \frac{1}{s} - \frac{A}{C}\frac{\hat{q}}{s\,\Theta_0} = \frac{1}{s} - \frac{A}{CD}\,\frac{E_dE\sigma^2 + \beta\sigma - \gamma}{\sigma^4(\sigma+c)},$$

using $Y_d + Y_s = D(\sigma + c)$ and $Y_dY_s = E_dE\sigma^2 + \beta\sigma - \gamma$. The inverse of $1/[\sigma^k(\sigma+c)]$, call it f_k, follows from $1/[\sigma^k(\sigma+c)] = c^{-1}\{\sigma^{-k} - 1/[\sigma^{k-1}(\sigma+c)]\}$. Starting from $f_1 = \mathrm{erfcx}(c\sqrt{t})$, this gives the recursion in Eq. (4.16). Numerically, $c\sqrt{t}$ is small, so evaluate the recursion with extra precision (or by its power series); the subtractions cancel otherwise.

6.2 The classical Stefan limit (sharp melting front)

If all the latent heat were absorbed at a sharp front at temperature T_f (true for plain ice water, with $T_f = 0\,°\mathrm{C}$), the liquid layer of thickness s around the can would carry heat quasi-steadily: $q = k_l(T_w - T_f)/[R\ln(1 + s/R)]$. Conservation of energy relates s to how far the drink has cooled. Including the sensible heat stored in the layer (mean excess about $\Theta/2$), with $\Theta = T_d - T_f$:

$$\Big(1 + \frac{s}{R}\Big)^2 = 1 + \frac{\beta\,(\Theta_0 - \Theta)}{1 + \gamma\Theta}, \qquad \beta = \frac{C}{\pi R^2 h\,\Delta H}, \qquad \gamma = \frac{C_v}{2\Delta H}.$$

For a well-mixed drink, $\mathrm{d}t = -(CR/Ak_l)\ln(1 + s/R)\,\mathrm{d}\Theta/\Theta$. The integral is two logarithms of linear functions divided by Θ, and each gives a dilogarithm Li_2:

$$t(\Theta) = \frac{CR}{2Ak_l}\Big[\ln(1+\beta\Theta_0)\ln\frac{\Theta_0}{\Theta} - \mathrm{Li}_2\frac{(\beta-\gamma)\Theta_0}{1+\beta\Theta_0} + \mathrm{Li}_2\frac{(\beta-\gamma)\Theta}{1+\beta\Theta_0} + \mathrm{Li}_2(-\gamma\Theta_0) - \mathrm{Li}_2(-\gamma\Theta)\Big].$$

(Verified by differentiating: sympy and a numerical quadrature agree to 10^{-10}.) Against the full solution this law is good for plain ice water (−6 %). Applied to salty slush with $T_f = T_L$, it is +18 % off at $\theta_b \approx 5\,\mathrm{K}$ and +57 % off at the eutectic, because it ignores the partly melted zone of §4.3. That failure is what led to the effusivity treatment.

6.3 Verification log

- **Algebra** (sympy): the regime-U quadratic (2.6), the eutectic and cotectic results (2.8)–(2.10), regime I (2.11), the eutectic boundary (2.12) and the dilogarithm antiderivative were each re-derived and checked by substitution.
- **Derivatives** (2.13): agree with finite differences of the closed form to four significant figures. **Inversion** (2.14): round trip exact. **Ethanol limit**: within 0.6 % of a brute-force root.

- **Bath**: 116 recipes against the earlier numerical model, worst 0.73 K, rms 0.27 K, one borderline regime call.
- **Full conduction solution**: halving the grid changes t_5 by under 0.1 s; energy is conserved to 0.3 %; the textbook Stefan problem is reproduced; a linear medium matches Eq. (4.10) within 1.1 %; a conducting drink in a linear medium matches Eq. (4.16) to 4 %.
- **Laplace inversions**: Eqs. (4.10) and (4.16) agree with a numerical inverse Laplace transform (Talbot's method, 30 digits) to 10^{-15}, for real roots, complex roots and the conducting drink. Eqs. (4.11) and (4.12) agree with numerical differentiation and quadrature of Eq. (4.10).
- **Closed form against the full solution**: 16 cases, rms 4.4 %, worst 10.7 %.

6.4 Sources

- R. J. Bodnar (1993), "Revised equation and table for determining the freezing point depression of H_2O–NaCl solutions", *Geochim. Cosmochim. Acta* 57, 683–684. NaCl liquidus to the eutectic.
- *Lange's Handbook of Chemistry* (ethanol–water freezing points); Å. Melinder (2010), *Properties of secondary working fluids for indirect systems*, IIR (via CoolProp INCOMP::MEA): ethanol–water freezing points and heat capacities.
- F. Farelo, A. M. C. Lopes, M. I. Ferra (2004), "Solubilities of sodium chloride and potassium chloride in water + ethanol mixtures from (298 to 323) K", *J. Chem. Eng. Data* 49, 1782–1788, doi:10.1021/je049782s. Salting-out of NaCl by ethanol.
- Excess enthalpies of ethanol + water at 298.15 K (the table used by the earlier numerical model; corrected to 0 °C with the excess heat capacity; **extraction to be checked against the paper**).
- R. H. Stokes, R. A. Robinson (1966), "Interactions in aqueous nonelectrolyte solutions. I. Solute–solvent equilibria", *J. Phys. Chem.* 70, 2126–2131; A. B. Zdanovskii (1936). The ZSR mixing rule.
- H. S. Carslaw, J. C. Jaeger (1959), *Conduction of Heat in Solids*, 2nd ed., Oxford: contact of two semi-infinite bodies; Neumann's two-region solution; small-time expansions for a cylinder.
- SINTEF can-cooling times (0.33 L can, 21 → 5 °C) and your measured can and tank dimensions: project notes `b8-web-sweep` and `b8-cooling-rig`.
- The earlier numerical bath model used for the comparison: `b8_core.py` (29 Oct 2026).